Spaceflight alters the tomato phyllosphere and rhizosphere microbiome in a light-dependent manner: a multi-omics integration of ISS VEG-05

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Abstract

Spaceflight imposes unique stressors on plant-microbe interactions, but the coordinated responses of host transcriptomes and associated microbiomes remain poorly characterized. We analyzed tomato (*Solanum lycopersicum* cv. Red Robin) grown aboard the International Space Station (ISS) as part of the VEG-05 experiment, integrating 16S rRNA and ITS amplicon sequencing (OSD-766) with host transcriptomics (OSD-767) across two tissues (leaf, adventitious root) and two light treatments (red-rich vs. blue-rich LED). DADA2 processing yielded 348 bacterial and 78 fungal ASVs from 142 and 141 samples respectively. Spaceflight accounted for 28% of the variation in leaf bacterial community composition (PERMANOVA, $\boldsymbol{p = 0.001}$), and flight samples had higher bacterial alpha diversity (Observed ASVs: Flight 17.8–25.8 vs. Ground 10.0–15.2). The spaceflight transcriptional response was strongly light-dependent: under blue light, 4,716 differentially expressed genes (DEGs) were detected in leaves versus only 523 under red light. WGCNA identified a 169-gene adventitious root module correlated with fungal dysbiosis ($\boldsymbol{r = -0.85}$, $\boldsymbol{p_{adj} = 0.009}$) but not with flight status ($\boldsymbol{r = -0.38}$, $\boldsymbol{p_{adj} = 0.32}$), representing a microbe-driven host transcriptional signature independent of the direct spaceflight stimulus. MOFA+ integration of 18 leaf samples across three omics layers found a dominant flight-associated factor (44% of transcriptome variance). The most abundant FAPROTAX-predicted bacterial functions were (aerobic) chemoheterotrophy, ureolysis, nitrogen fixation and methylotrophy. These findings demonstrate that spaceflight restructures plant-associated microbiomes in a tissue- and light-specific manner, with implications for crop production systems in long-duration space missions.

Keywords: spaceflight, plant microbiome, multi-omics, VEG-05, tomato, WGCNA, MOFA+, dysbiosis

1 Introduction

Plant-associated microbiomes play critical roles in nutrient acquisition, pathogen protection, and stress tolerance [1]. In the context of spaceflight, these interactions take on added significance: crops grown in bioregenerative life support systems must maintain productive plant-microbe partnerships under conditions of microgravity, altered fluid dynamics, elevated radiation, and controlled-environment lighting [**TODO: citation needed**]. The VEG-05 experiment aboard the International Space Station (ISS) provided a unique opportunity to study these interactions in tomato (*Solanum lycopersicum*), a candidate crop for long-duration missions [2, 3].

Spaceflight has been shown to alter microbial community composition, gene expression, and virulence in pure culture systems [**TODO: citation needed**]. However, the coordinated response of host plants and their associated microbiomes to the spaceflight environment remains largely unexplored. Multi-omics integration approaches such as MOFA+ (Multi-Omics Factor Analysis) [4] and WGCNA (Weighted Gene Co-expression Network Analysis) [5] offer powerful frameworks for identifying coordinated variation across biological layers, but have not been applied to spaceflight plant-microbiome studies.

Here we present an integrated analysis of the VEG-05 tomato microbiome (16S rRNA and ITS amplicon sequencing, OSD-766) and host transcriptome (RNA-seq, OSD-767) (Fig. 1). We characterize community health through diversity metrics and a dysbiosis index, identify co-expression modules associated with microbial changes, and infer the directionality of host-microbe interactions using a framework that distinguishes host-driven, microbe-driven, and co-regulated responses.

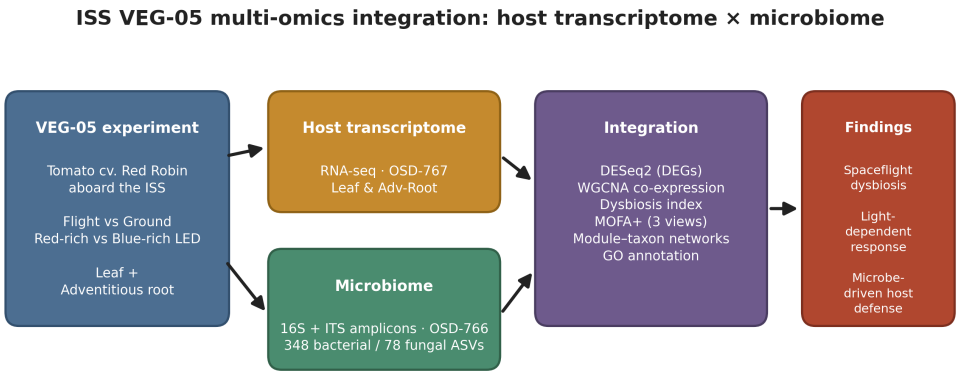


Fig. 1 Study design and analysis overview. The ISS VEG-05 experiment (tomato cv. Red Robin; Flight vs Ground; red-rich vs blue-rich LED; leaf and adventitious root) feeds two data streams — host transcriptome (OSD-767) and microbiome (OSD-766) — combined through DESeq2, WGCNA, a dysbiosis index, MOFA+, and module–taxon networks.

2 Methods

2.1 Data acquisition and metadata

Raw sequencing data and metadata were obtained from NASA's Open Science Data Repository (OSDR): OSD-766 (microbiome amplicon sequencing; [2]) and OSD-767 (host RNA-seq; [3]). The VEG-05 experiment grew tomato cv. Red Robin aboard the ISS under red-rich (90% red:10% blue) or blue-rich (50% red:50% blue) LED lighting, with ground controls at Kennedy Space Center. Metadata were parsed from ISA-Tab format, and a crosswalk table linking microbiome and RNA-seq samples by plant ID, flight status, light treatment, and tissue was constructed. OSD-766 labels the adventitious-root samples of ground plants SN02, SN04 and SN05 blue-rich, whereas it labels those plants' leaf samples red-rich and OSD-767 names their root RNA-seq samples Red; these three samples were treated as red-rich.

2.2 Microbiome amplicon processing

16S rRNA (V4 region, 515F/806R primers) and ITS2 (ITS3F/ITS4 primers) reads were processed using DADA2 v1.40.0 [6] in R. For 16S, primers were already removed; reads were truncated at 240 bp (R1) and 160 bp (R2) with maxEE = 2, length-filtered to 200–260 bp, and pooled for denoising. Taxonomy was assigned against SILVA 138.1 [7]. Chloroplast and mitochondrial ASVs were removed prior to community analysis. For ITS, primers were removed with cutadapt [8]; R2 reads were truncated at 180 bp due to quality degradation; taxonomy was assigned against the UNITE v7.2 QIIME release (1 December 2017, 99% clusters) [9]. Taxonomy was assigned with a fixed random seed and DADA2's default 50% bootstrap threshold.

2.3 Differential expression analysis

RNA-seq count matrices were analyzed with DESeq2 [10] separately for leaf (21 samples) and adventitious root (15 samples; Table S1). The design model included flight status, light treatment, and their interaction. Each main effect is the effect of one factor averaged over both levels of the other. Log-fold changes were shrunk with ashr [11] for the main effects and the within-light contrasts, and with apeglm [12] for the interaction term. DEGs were defined as $p_{\text{adj}} < 0.05$ and $|\log2\text{FC}| \geq 1$.

2.4 Community health metrics

Alpha diversity (Observed ASVs, Shannon index) was computed on unfiltered data using phyloseq [13]. Beta diversity was assessed via Bray-Curtis dissimilarity and PERMANOVA (999 permutations; terms added sequentially: flight, light, flight × light) using vegan [14]. A dysbiosis index was calculated for each flight sample as its mean Bray-Curtis distance to all ground samples of the same compartment (both light treatments), divided by the mean pairwise distance among those ground samples. Differential abundance testing used ALDEx2 [15] with 128 Monte Carlo instances, requiring both a Benjamini–Hochberg-adjusted Welch $p < 0.05$ and $|\text{effect size}| > 1$; positive effects are higher in flight.

2.5 WGCNA co-expression networks

WGCNA [5] was applied to variance-stabilized transcriptome data for each tissue separately. Soft-thresholding powers were selected to maximize the signed scale-free fit, as neither tissue reached the 0.8 target (Leaf: power = 20, $R^2 = 0.72$; Adv-Root: power = 18, $R^2 = 0.69$; Fig. S7). Module eigengenes were correlated (Pearson) with flight status, light treatment, and the 16S/ITS dysbiosis indices of the same plant and tissue. A ground sample's index is its mean Bray-Curtis distance to the other ground samples of its compartment, over the same denominator, so ground indices average 1. Dysbiosis indices exist for 21 (16S) and 18 (ITS) of the 21 leaf samples and for 15 and 12 of the 15 adventitious-root samples; each correlation's p-value uses that number of samples. Hub genes were the 10 genes with the highest module membership (k_{ME}) in each module (all $k_{\mathrm{ME}} \geq 0.85$).

2.6 MOFA+ multi-omics integration

MOFA2 [4] was trained on the 18 leaf samples with RNA-seq, 16S and ITS data from the same leaf, in three views: transcriptome (2,000 most variable genes), 16S (the 101 bacterial ASVs detected in those samples), and ITS (40 fungal ASVs). The model was trained with 5 factors and default priors. Factor-trait correlations were computed using Spearman's rank correlation with BH correction.

2.7 Correlation networks and directionality inference

Bipartite module-taxon networks were constructed by Spearman correlation between WGCNA module eigengenes and ASV relative abundances (prevalence filter $> 20\%$, BH correction). GO enrichment of WGCNA modules was performed using clusterProfiler's enricher function [16] with tomato GO annotations from Ensembl Plants [**TODO: Ensembl Plants release and number of GO-annotated genes**]. Directionality was classified as: host-driven (correlated with flight/light but not dysbiosis), microbe-driven (correlated with dysbiosis but not flight/light), or co-regulated (correlated with both).

2.8 Functional prediction

Bacterial functional profiles were predicted for the 348 bacterial ASVs using FAPROTAX 1.2.12 [17] and its collapse_table.py (whole-word taxon matching, with hyphens treated as word separators so that each genus in SILVA's merged genus names is matched). Fungal ecological guilds were assigned using genus-level lookup tables based on FUNGuild [18] categories.

3 Results

3.1 Spaceflight reshapes leaf bacterial communities

DADA2 processing of 16S rRNA reads yielded 544 ASVs from 142 samples (3,558,814 non-chimeric reads, 89.8% retention). After removing chloroplast (38.9% of reads)

and mitochondrial (21.5%) sequences, 348 bacterial ASVs remained. ITS processing yielded 78 fungal ASVs (156,102 reads, 15.4% of primer-trimmed reads); 121 of the 141 samples retained reads.

Spaceflight changed bacterial community composition in leaves (PERMANOVA flight term: $R^2 = 0.28$, $F = 8.24$, $p = 0.001$, $n = 21$) and adventitious roots ($R^2 = 0.29$, $F = 5.90$, $p = 0.001$, $n = 15$; Table S4). Light quality had no significant main effect in either tissue, but the flight $\times$ light interaction did (leaf $R^2 = 0.11$, $p = 0.035$; root $R^2 = 0.11$, $p = 0.027$). In leaves, bacterial alpha diversity was higher in flight samples: Observed ASVs ranged from 17.8–25.8 in flight versus 10.0–15.2 in ground controls (Fig. 2; Table S3). The Shannon index showed a similar pattern, with flight blue-light samples reaching 2.18 versus 0.80 in ground blue-light controls.

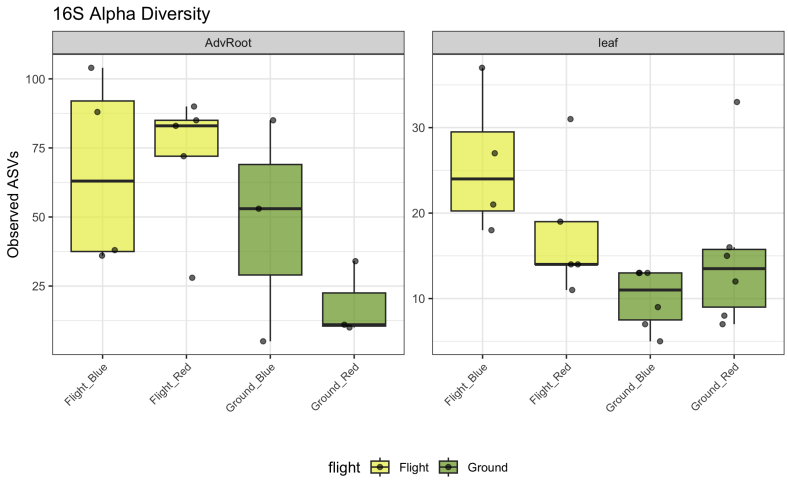


Fig. 2 Bacterial (16S) alpha diversity (Observed ASVs) by flight status and light treatment; flight leaves show higher diversity.

The dysbiosis index quantified community displacement relative to ground controls. Leaf bacterial communities showed a 2.06-fold increase in dysbiosis during flight (2.06$\pm$ 0.99 vs. 1.00 $\pm$ 0.47; Fig. 3). Fungal communities showed even greater displacement (ITS leaf: 3.69 $\pm$ 2.78 vs. 1.00 $\pm$ 0.22; Fig. 3), and spaceflight changed leaf fungal composition (flight term $R^2 = 0.36$, $F = 9.35$, $p = 0.007$, $n = 18$). Adventitious root fungal community composition differed by flight ($R^2 = 0.40$, $F = 13.1$, $p = 0.001$), light ($R^2 = 0.19$, $p = 0.005$) and their interaction ($R^2 = 0.16$, $p = 0.006$; $n = 12$).

ALDEx2 identified few taxa with significant differential abundance under the conservative dual criterion (BH-adjusted $p < 0.05$ and $|\text{effect}| > 1$), all of them more abundant in flight. In leaf 16S, an Erwiniaceae ASV (ASV024; assigned to *Pantoea* with 49% bootstrap support, just below the 50% threshold) was enriched in flight (effect = +1.10). In ITS, *Penicillium citrinum* (ASV003) was enriched in flight leaves (effect = +1.90). In adventitious roots, three bacterial taxa (16S ASV004, *Burkholderia-Caballeronia-Paraburkholderia*, +1.58; ASV021, Rhizobiaceae, +3.25;

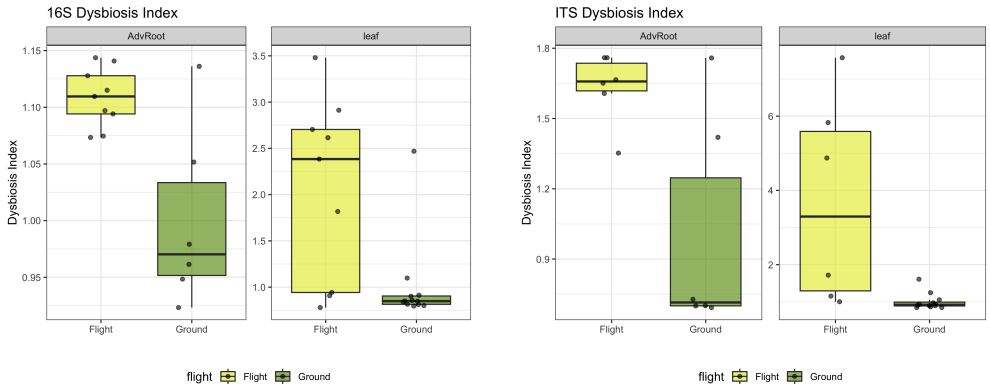


Fig. 3 Dysbiosis index (mean Bray-Curtis distance to ground samples, relative to ground-ground distance). Left (16S): bacterial dysbiosis, elevated under flight. Right (ITS): fungal dysbiosis, largest displacement in flight leaves.

ASV022, *Dyadobacter*, +2.22) were enriched in flight. A *Fusarium* ASV (ITS ASV006) had a similar effect in roots (+1.91) but did not pass the threshold (BH-adjusted $p = 0.053$).

3.2 Spaceflight transcriptional response is light-dependent

DESeq2 analysis revealed a pronounced interaction between spaceflight and light quality (Table S2). In leaves, the flight main effect (the flight effect averaged over the two light treatments) gave 2,683 DEGs (1,368 up, 1,315 down; Fig. 5). Separate models within each light treatment gave 4,716 DEGs under blue-rich light and 523 under red-rich light (Fig. 4). The formal interaction term identified 3,189 DEGs, confirming that the transcriptional response to spaceflight is strongly modulated by light quality.

In adventitious roots, the flight main effect was predominantly upregulation (1,501 DEGs: 1,245 up, 256 down, 83% upregulated; Fig. 5), suggesting activation of stress response pathways. The interaction was minimal (6 DEGs after shrinkage), indicating that the root transcriptional response to spaceflight is less light-dependent than the leaf response.

3.3 WGCNA identifies a microbe-driven host transcriptional module

WGCNA of the leaf transcriptome identified 12 modules plus unassigned genes (power = 20; Fig. 6), with the turquoise module ($n = 2{,}037$ genes) showing the strongest flight correlation ($r = +0.73$, $p_{\text{adj}} = 0.001$) and the blue module ($n = 1{,}012$) the strongest anti-correlation ($r = -0.79$, $p_{\text{adj}} = 3 \times 10^{-4}$). Four leaf modules were classified as co-regulated (correlated with both flight and dysbiosis; Table S5).

In adventitious roots, WGCNA identified 16 modules plus unassigned genes (power = 18). The key finding was the black module ($n = 169$ genes), strongly correlated with ITS dysbiosis ($r = -0.85$, $p_{\text{adj}} = 0.009$, 12 plants) but not with flight status ($r = -0.38$, $p_{\text{adj}} = 0.32$; Fig. 6). The correlation holds within flight plants ($r = -0.87$,

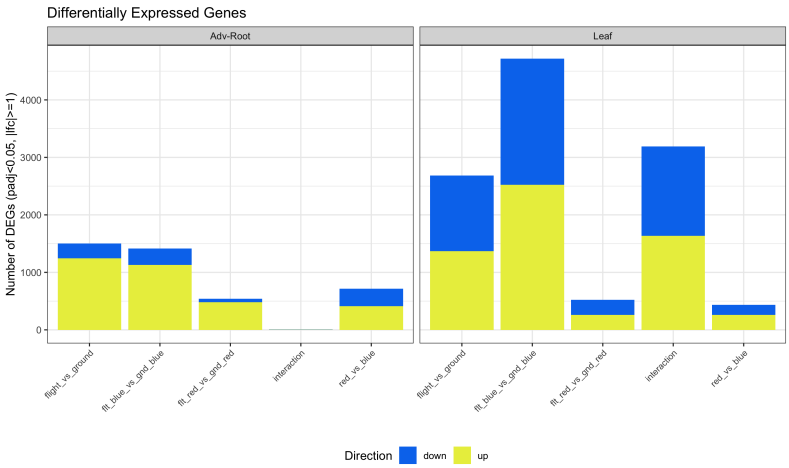


Fig. 4 DEG summary across tissue × contrast; the leaf blue-light flight response (4,716 DEGs) dwarfs red (523).

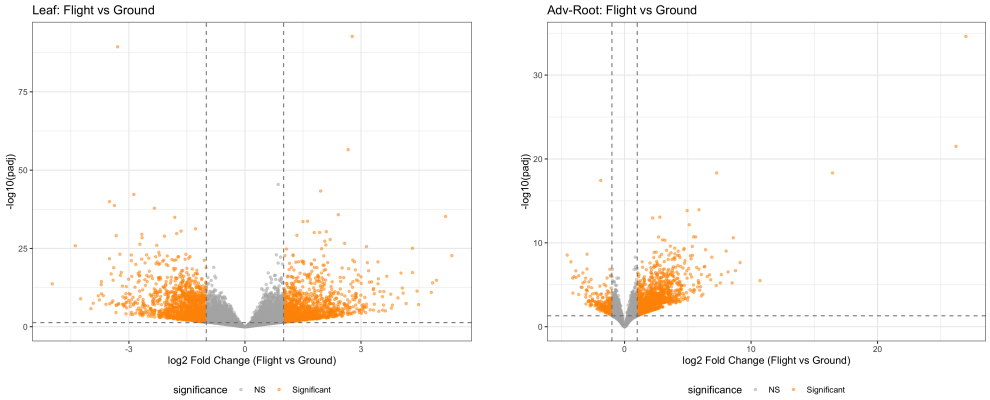


Fig. 5 Flight-vs-Ground volcano plots for the flight main effect of the flight × light model, i.e. the flight effect averaged over the two light treatments. Left (3*B*): leaf. Right (3*C*): adventitious root (predominantly upregulation).

$n = 6$) and within ground plants ($r = -0.75$, $n = 6$), although the ITS dysbiosis index is itself higher in flight ($r = 0.71$ with flight status). The green module ($n = 287$; ITS $r = -0.72$, $p_{\text{adj}} = 0.044$) was also classified as microbe-driven, and the turquoise module as co-regulated (flight $r = +0.72$; ITS $r = +0.74$, $p_{\text{adj}} = 0.044$). The black module was classified as microbe-driven, representing a host transcriptional response to fungal community changes independent of the direct spaceflight stimulus. GO enrichment of the black module itself was limited — its only significantly enriched term was the cellular component "nucleus" — so it is defined here by its microbe-driven trait association; the adventitious-root oxidative-stress program was instead concentrated in the larger turquoise module (see below).

GO enrichment of all modules revealed biologically coherent signatures (Table S10). In leaves, the black module was enriched for photosynthesis and photosystem genes ($p_{\text{adj}} = 4.4 \times 10^{-25}$), while the red module was enriched for RNA binding and protein binding. In adventitious roots, the turquoise module (the largest, $n = 1{,}768$) was enriched for oxidative stress response (peroxidase activity, $p_{\text{adj}} = 4.2 \times 10^{-9}$; hydrogen peroxide catabolism, $p_{\text{adj}} = 1.1 \times 10^{-8}$; glutathione transferase activity, $p_{\text{adj}} = 3.9 \times 10^{-7}$), consistent with spaceflight-induced oxidative stress.

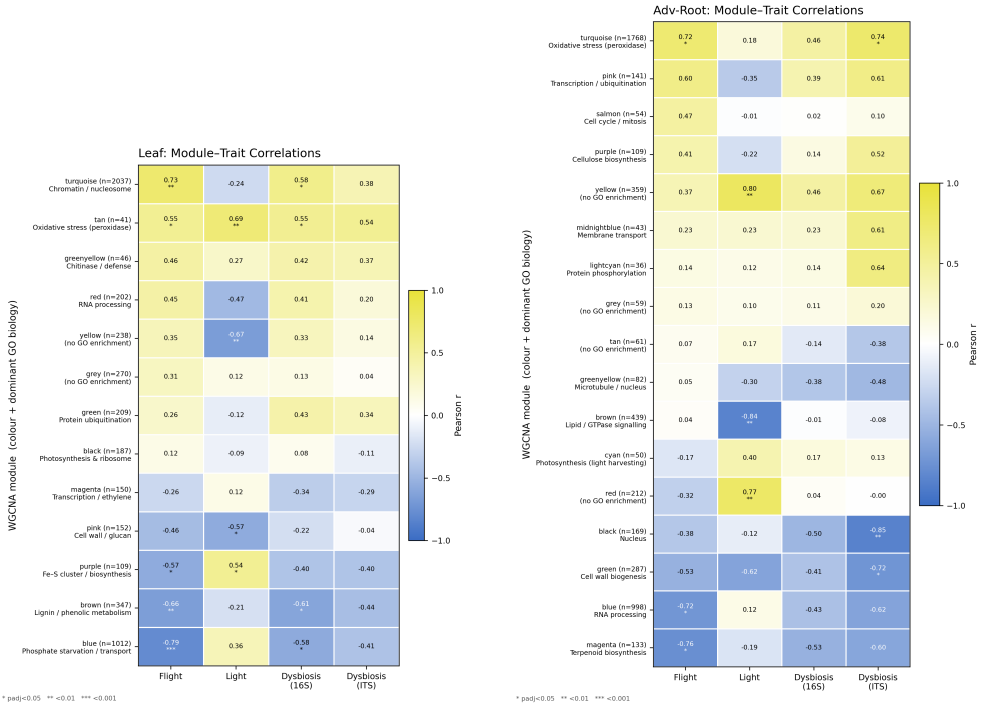


Fig. 6 WGCNA module–trait correlations (flight, light, 16S/ITS dysbiosis). Left: leaf. Right: adventitious root — the black (nucleus) module tracks ITS dysbiosis ($r = -0.85$) but not flight; the turquoise (oxidative-stress/peroxidase) module is the largest flight-correlated module. Each module is labelled with its dominant GO biology (Table 1).

3.4 Module functional annotation (GO)

To make the WGCNA module colours interpretable, each module was annotated with its dominant biology from per-module Gene Ontology enrichment (Table 1). These human-readable labels are shown alongside the module colours on the axes of Figs 6 and 8. Modules with no significant enrichment are marked accordingly.

Table 1: GO-based functional annotation of WGCNA modules.

Tissue / Module	GO biology (dominant)	Representative GO terms
Adv-Root / black	Nucleus	nucleus
Adv-Root / blue	RNA processing	protein binding; RNA binding; RNA modification
Adv-Root / brown	Lipid / GTPase signalling	ATP-dependent diacylglycerol kinase activity; phospholipase C-activati...
Adv-Root / cyan	Photosynthesis (light harvesting)	photosynthesis, light harvesting; photosynthesis; photosystem I
Adv-Root / green	Cell wall biogenesis	xyloglucan metabolic process; cell wall; xyloglucan:xyloglucosyl trans...
Adv-Root / greenyellow	Microtubule / nucleus	microtubule binding; ATP binding; nucleus
Adv-Root / lightcyan	Protein phosphorylation	protein phosphorylation; protein kinase activity; ATP binding
Adv-Root / magenta	Terpenoid biosynthesis	acyltransferase activity; terpene synthase activity; terpenoid biosynt...
Adv-Root / midnightblue	Membrane transport	transmembrane transporter activity; transmembrane transport; membrane
Adv-Root / pink	Transcription / ubiquitination	DNA-binding transcription factor activity; ubiquitin protein ligase ac...
Adv-Root / purple	Cellulose biosynthesis	cellulose synthase (UDP-forming) activity; cellulose biosynthetic proc...
Adv-Root / salmon	Cell cycle / mitosis	cyclin-dependent protein serine/threonine kinase regulator activity; m...
Adv-Root / turquoise	Oxidative stress (peroxidase)	peroxidase activity; hydrogen peroxide catabolic process; heme binding
Leaf / black	Photosynthesis & ribosome	photosynthesis; ribosome; photosystem I
Leaf / blue	Phosphate starvation / transport	DNA-binding transcription factor activity; cellular response to phosph...
Leaf / brown	Lignin / phenolic metabolism	heme binding; endopeptidase inhibitor activity; cation binding
Leaf / green	Protein ubiquitination	protein ubiquitination; ubiquitin-protein transferase activity; cell s...
Leaf / greenyellow	Chitinase / defense	chitinase activity; chitin catabolic process; chitin binding

Table 1 — continued

Tissue / Module	GO biology (dominant)	Representative GO terms
Leaf / magenta	Transcription / ethylene	regulation of DNA-templated transcription; ethylene-activated signalin…
Leaf / pink	Cell wall / glucan	cell wall; xyloglucan:xyloglucosyl transferase activity; glucan metabo…
Leaf / purple	Fe–S cluster / biosynthesis	iron-sulfur cluster binding; biosynthetic process
Leaf / red	RNA processing	RNA binding; protein binding; RNA modification
Leaf / tan	Oxidative stress (peroxidase)	hydrogen peroxide catabolic process; peroxidase activity; response to…
Leaf / turquoise	Chromatin / nucleosome	nucleosome; structural constituent of chromatin; microtubule binding

3.5 MOFA+ confirms dominant flight factor and cross-omics coordination

MOFA+ integration of the 18 leaf samples with RNA-seq, 16S and ITS data from the same leaf (transcriptome: 2,000 genes; 16S: 101 ASVs; ITS: 40 ASVs) identified 5 factors. Factor 1 was the dominant signal, explaining 44.4% of transcriptome variance, 9.0% of 16S variance, and 7.0% of ITS variance (Fig. 7). Factor 1 was significantly correlated with flight status (Spearman $\rho = -0.68$, $p_{\text{adj}} = 0.037$; Fig. 7) and nearly as strongly with light treatment ($\rho = +0.63$, $p_{\text{adj}} = 0.053$); flight and light are partly confounded in these samples, because only one flight blue-light leaf has ITS data. Factor 3 trended with 16S dysbiosis ($\rho = -0.61$, $p = 0.008$, $p_{\text{adj}} = 0.053$); no other factor–trait correlation had $p_{\text{adj}} < 0.1$. The transcriptome contributed the highest-weighted features to Factors 1–4 and 16S ASVs to Factor 5 (Fig. S6).

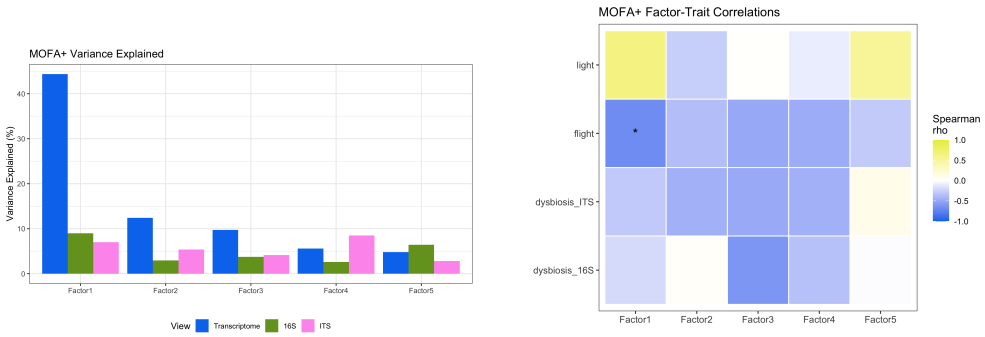


Fig. 7 MOFA+ integration. Left (5*A*): variance explained per factor per view. Right (5*B*): factor–trait correlations; Factor 1 captures flight ($\rho = -0.68$).

3.6 Module-taxon correlations link host modules to specific bacteria

Bipartite correlation networks identified 31 significant module-taxon relationships (BH $p_{\text{adj}} < 0.05$), predominantly in leaf 16S (28 of 31; Fig. 8; Table S9). The other three link adventitious-root modules to fungal ASVs, two of them to a *Fusarium* ASV (ITS ASV006; blue module $\rho = -0.93$, turquoise $\rho = +0.91$). The flight-associated turquoise module positively correlated with *Methylobacterium-Methylorubrum* ($\rho = +0.76$, $p_{\text{adj}} = 0.003$), *Burkholderia-Caballeronia-Paraburkholderia* ($\rho = +0.74$, $p_{\text{adj}} = 0.003$), and *Azospirillum* ($\rho = +0.68$, $p_{\text{adj}} = 0.012$) — all genera associated with plant growth promotion and nitrogen fixation. The anti-correlated blue module showed the inverse pattern. The brown module correlated negatively with the flight-enriched Erwiniaceae ASV (ASV024; $\rho = -0.79$) and *Paenibacillus* ($\rho = -0.77$).

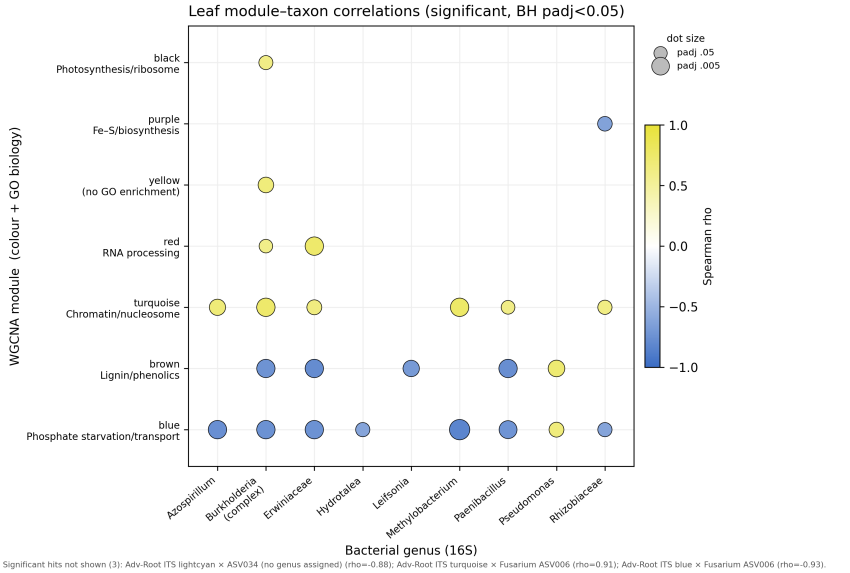


Fig. 8 Leaf module–taxon correlations (BH $p_{\text{adj}} < 0.05$). The flight-associated turquoise (chromatin/nucleosome) module positively tracks *Methylobacterium*, *Burkholderia*, and *Azospirillum*, while the blue (phosphate-starvation) module shows the inverse.

3.7 Predicted bacterial functions

FAPROTAX assigned 205 of the 348 bacterial ASVs to 44 functional groups (Table S6). By read count the largest were chemoheterotrophy (786,147 reads, 180 ASVs) and its subset aerobic chemoheterotrophy (688,214 reads, 150 ASVs), ureolysis (235,475 reads, 31 ASVs), nitrogen fixation (223,554 reads, 15 ASVs), methylotrophy (96,868 reads, 16 ASVs) and methanol oxidation (96,687 reads, 15 ASVs) (Fig. 9). The methylotrophy and methanol oxidation assignments were almost entirely *Methylobacterium-Methylorubrum* (14 ASVs), the genus most strongly correlated with

the flight-associated turquoise module. Methanotrophy was minor (181 reads, 1 ASV): *Methylobacterium* oxidizes methanol, not methane. Fungal guild assignment (manual genus-level classification) identified saprotrophs (11 ASVs, dominated by *Penicillium* and *Aspergillus*) and plant pathogens (2 ASVs: *Fusarium* spp.) as the main guilds (Table S7).

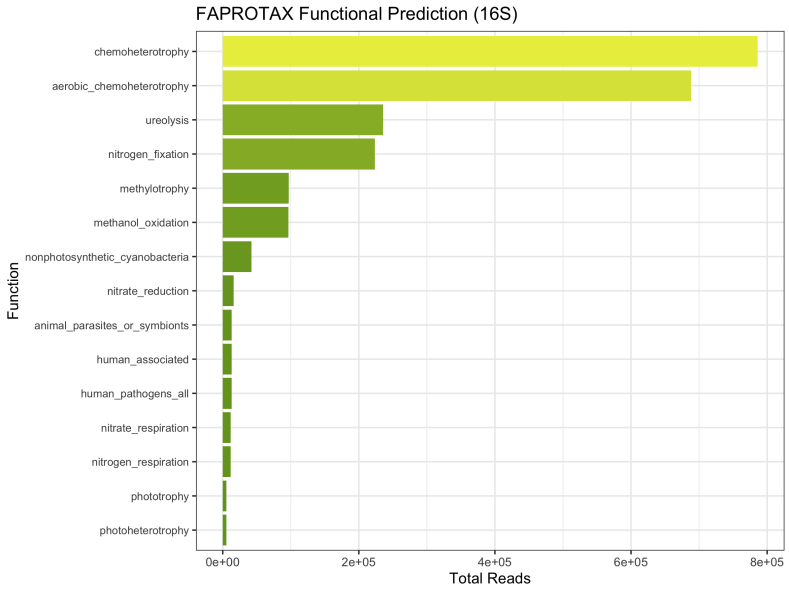


Fig. 9 FAPROTAX predicted bacterial functions (FAPROTAX 1.2.12; the 15 largest groups by read count: chemoheterotrophy, ureolysis, nitrogen fixation, methylotrophy and others).

4 Discussion

This study presents the first integrated multi-omics analysis of plant-microbiome interactions in the ISS VEG-05 experiment. Our findings reveal three key insights: (1) spaceflight restructures both the bacterial and the fungal communities of leaves, with increased bacterial alpha diversity in flight; (2) the host transcriptional response to spaceflight is strongly light-dependent, particularly in leaves; and (3) a subset of host genes responds to microbial community changes independently of the direct spaceflight stimulus, providing evidence for microbe-driven host regulation.

The increased bacterial alpha diversity in flight leaves was initially counterintuitive, as spaceflight is often associated with reduced microbial diversity in human microbiome studies [19]. However, in the phyllosphere, the reduced gravity environment may alter leaf surface properties and nutrient availability, potentially creating more ecological niches. The strong correlation between the flight-associated WGCNA module and plant growth-promoting bacteria (*Methylobacterium*, *Burkholderia*, *Azospirillum*)

suggests that spaceflight may favor these beneficial taxa, though the directionality of this association requires further investigation.

The light-dependent transcriptional response is a novel finding with practical implications for space crop production. The 9-fold difference in DEG counts between blue and red light conditions (4,716 vs. 523) suggests that blue light may amplify the plant's perception of or response to the spaceflight environment. Blue light is known to activate cryptochrome photoreceptors and downstream stress signaling pathways [20], which may interact with spaceflight-induced oxidative stress.

The identification of a microbe-driven WGCNA module in adventitious roots (black module, 169 genes) is perhaps the most significant finding. This module's strong correlation with fungal dysbiosis ($r = -0.85$) but not with flight status ($r = -0.38$, $p_{\text{adj}} = 0.32$) indicates that the host transcriptional response in this module is driven by microbial community changes rather than directly by the spaceflight environment. Although the black module's own GO enrichment was limited, the broader adventitious-root oxidative-stress program — concentrated in the larger turquoise module — together with the black module's tight coupling to fungal dysbiosis suggests that fungal community restructuring may trigger host defense and redox responses, potentially through oxidative burst signaling.

4.1 Relationship to a cross-mission salicylic-acid integration

These microbiome findings dovetail with a companion cross-mission analysis that integrated the same OSD-767 transcriptome with an Advanced Plant Habitat experiment manipulating salicylic-acid (SA) signalling (wild-type MoneyMaker versus the SA-deficient *NahG* line). That analysis concluded that the spaceflight transcriptome resembles an SA-deficient, biotic-defense-primed state. Our results offer a proximal explanation: spaceflight-induced dysbiosis — and the microbe-driven, flight-independent host module identified here — could be the stimulus that elicits that defense-primed signature, with SA acting as the gating regulator of plant biotic immunity. Both analyses independently recover the blue-light amplification of the spaceflight response despite using different pipelines, and this rationalises why gene-level spaceflight signatures replicate poorly across missions while the biotic-defense *pattern* is conserved. A full comparison is provided in `COMPARISON_with_SA_integration.md`.

4.2 Limitations

Several limitations should be noted. First, sample sizes are small ($n = 21$ leaf, $n = 15$ root for RNA-seq; $n = 1$–6 per flight $\times$ light group for microbiome), limiting statistical power. Second, 16S leaf samples had very low bacterial read counts after host DNA removal (median 89 reads), precluding rarefaction and requiring reliance on relative abundance metrics. Third, the ITS dataset had uneven group representation, with some flight groups containing only 1–2 samples, and two flight adventitious-root samples retained a single ITS read each. Fourth, MOFA+ warned that 18 samples are few for five factors (it suggests at most about four), and that Factor 1 is strongly correlated with the average level of the features in at least one view, which happens when samples differ in overall levels, sometimes through normalisation; Factor

1's correlation with flight ($\rho = -0.68$) does not rule this out. Fifth, FAPROTAX functional predictions are taxonomy-based and should be validated with metagenomic or metatranscriptomic data. Sixth, the FUNGuild analysis used manual genus-level assignment rather than the full database, reducing assignment confidence. Finally, this study is correlational; causal relationships between host transcription and microbiome changes require experimental validation, ideally in simulated microgravity conditions with microbiome manipulation.

Supplementary figure legends

Figure S1. DADA2 read retention. Reads per sample at each DADA2 step (input, filtered, denoised forward and reverse, merged, non-chimeric) for all 142 16S and 141 ITS samples, on a log scale. Samples with no reads at a step cannot be drawn on that scale; 20 ITS samples have no non-chimeric reads.

Figure S2. Alpha diversity in every compartment. Observed ASVs, Shannon and Simpson indices for leaf, adventitious root (AdvRoot), root, wick and soil samples, flight and ground, for 16S (78 samples; chloroplast and mitochondrial ASVs removed) and ITS (65 samples). Points are jittered sideways only.

Figure S3. Bray–Curtis PCoA. Principal coordinates of Bray–Curtis dissimilarities between relative-abundance profiles for 16S (78 samples; chloroplast and mitochondrial ASVs removed) and ITS (65 samples), coloured by compartment, with flight and ground as point shapes. Ellipses are 95% normal-theory ellipses per compartment; axis percentages are each axis's share of the eigenvalue sum.

Figure S4. Bacterial genera by compartment (16S). Mean relative abundance per compartment of the 15 genera with the highest mean over the five compartments, after removing chloroplast and mitochondrial ASVs; all other genera are pooled as Other. ASVs without a genus call are grouped by family (Unclassified <family>).

Figure S5. All nominal module–taxon correlations. Spearman correlations between WGCNA module eigengenes and ASV relative abundances with unadjusted $p < 0.05$ (449 correlations), by tissue and amplicon. Rows are genera, or the family where no genus was assigned; ASVs with neither are pooled. Where several ASVs of a row correlate with one module, the tile shows the strongest (302 tiles). Correlations that pass BH correction are shown in Fig. 6 and listed in Table S9.

Figure S6. MOFA+ feature weights. The 20 features with the largest absolute weight on each factor in each view (transcriptome genes, 16S and ITS ASVs) of the MOFA+ model in Fig. 5 (18 leaf samples).

Figure S7. WGCNA soft-threshold selection. Signed scale-free topology fit (WGCNA pickSoftThreshold, signed network: $-\text{sign}(\text{slope}) \times R^2$) against soft power for the leaf (21 samples) and adventitious-root (15 samples) expression matrices of 5,000 genes each. The dashed line is the 0.8 target, which neither tissue reached; the power with the highest fit was used (dotted lines: leaf 20, $R^2 = 0.72$; adventitious root 18, $R^2 = 0.69$).

Data and code availability. All analysis code is available at https://github.com/dr-richard-barker/veg05-integrated-omics. Raw data are available from NASA OSDR (OSD-766, OSD-767).

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